

INTRODUCTION

Natural language processing

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Vision & Mission of AIUB

- **Vision**

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB) envisions promoting professionals and excellent leadership catering to the technological progress and development needs of the country.

- **Mission:**

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB) is committed to provide quality and excellent computer-based academic programs responsive to the emerging challenges of the time. It is dedicated to nurture and produce competent world class professional imbued with strong sense of ethical values ready to face the competitive world of arts, business, science, social science and technology.

Goals of AIUB

- Sustain development and progress of the university
- Continue to upgrade educational services and facilities responsive of the demands for change and needs of the society
- Inculcate professional culture among management, faculty and personnel in the attainment of the institution's vision, mission and goals
- Enhance research consciousness in discovering new dimensions for curriculum development and enrichment
- Implement meaningful and relevant community outreach programs reflective of the available resources and expertise of the university
- Establish strong networking of programs, sharing of resources and expertise with local and international educational institutions and organizations
- Accelerate the participation of alumni, students and professionals in the implementation of educational programs and development of projects designed to expand and improve global academic standards

Vision & Mission of CS Department

- **Vision**

Provides leadership in the pursuit of quality and excellent computer education and produce highly skilled and globally competitive IT professionals.

- **Mission**

Committed to educate students to think analytically and communicate effectively; train them to acquire technological, industry and research-oriented accepted skills; keep them abreast of the new trends and progress in the world of information communication technology; and inculcate in them the value of professional ethics.

Goals of CS Department

- Enrich the computer education curriculum to suit the needs of the industry-wide standards for both domestic and international markets
- Equip the faculty and staff with professional, modern technological and research skills
- Upgrade continuously computer hardware, facilities and instructional materials to cope with the challenges of the information technology age
- Initiate and conduct relevant research, software development and outreach services.
- Establish linkage with industry and other IT-based organizations/institutions for sharing of resources and expertise, and better job opportunities for students

Semester Schedule (Summer 25-26)

- Week 1 - Week 6: Classes
- Week 7: Midterm Exam
- Week 8 – Week 13: Classes
- Week 14: Final Exam

Course Evaluation

Mid Term	Attendance	10	40%
	Quizzes	30	
	Tasks/Presentation	10	
	Assignment + Viva	50	
Final Term	Attendance	10	60%
	Quizzes	30	
	Tasks/Presentation	10	
	Project Implementation + VIVA	50	
Grand			100%

Course Requirements

- **Must have at least 80% attendance to pass this course.** Failure to do so will result in an **automatic UW**.
- Must appear in the midterm and final term exam.
- No make-up Quiz.
- No request will be entertained for raising grade.

Attendance

- If your attendance falls below 75%, you will get UW.
- If you are absent for some reason, then you must submit an application (hard copy, in the given format) with valid supporting documents when you come next time in the class.
- If you are in probation, do NOT miss any class.
- **Note**: A Probation student can NOT DROP course.

Cheating Policy (Official Rule)

- AIUB does not allow any kind of cheating.
- In case of cheating, you'll get an **F** in the course.

Teaching

- Class lectures are mandatory.
 - Class attendance
 - Late in class
 - Class break
- Course materials will be provided (on VUES) and suggested time to time.
- Consultation /Communication
- Grading: F/ I /UW

Course Outline

This course focuses on learning NLP techniques.

- NLP Basics
- Regular Expression
- Text Processing/Normalization
- Vector Semantics and Embeddings
- Text Classification
- Edit Distance
- N-Gram Language Models
- Neural Networks
- Transformers

Text/Reference Books

- Daniel Jurafsky and James H. Martin. 2025. **Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models**, 3rd edition.
- Hobson Lane, Maria Dyshel. 2025. Natural Language Processing in Action.

Questions?

Thank you!